Module 3 Summary

Response to feedback

If you are resubmitting, include a statement outlining the changes you have made to your submission. This section can be short but should be precise. It is a good idea to quote the feedback you are responding to.

If this is your first submission, include a statement about what part of the lesson review you would most like to receive feedback (and why). Your tutor will take this into consideration when reviewing your work, although they may choose to give you feedback on a different thing if they think it's more appropriate.

Module Learning Objectives

I certify that I achieved the following learning objectives for the module (these objectives can be found in the introduction of the module):

- …
- …

Key Definitions and Theorems

Matrix Inverses

Textbook Ref	Theorem/Definition statement	Learning Notes
Definition 2.11 Matrix Inverse	If A is a square matrix, a matrix B is called an <u>inverse</u> of A if and only if $AB = I$ and $BA = I$ A matrix A that has an inverse is called an <u>invertible matrix</u> and denoted A^{-1}	DIFFERENT FROM THE NEGATION, remember that this isn't the negative matrix but its inversion
Theorem 2.4.1 Uniqueness of inverse	If B and C are both inverses of A, then B = C.	Think about the rules with matrix multiplication, sometimes C might not be known but must still be identical if it returns the same result as B, there's a single inverse for every invertible matrix
Matrix Inversion Algorithm	If A is an invertible (square) matrix, there exists a sequence of elementary row operations that carry A to the identity matrix I of the same size, written $A \to I$. This same series of row operations carries I to A^{-1}; that is, $I \to A^{-1}$. The algorithm can be	Note that the inverse doesn't always exist, and REMEMBER that the identity matrix is the matrix representation of scalar 1, it leaves it

	summarized as follows: $[A \; I] \rightarrow [I \; A^{-1}]$ where the row operations on A and I are carried out simultaneously.	unchanged (replacing the 1s with any real x will be that scalar's matrix representation, I assume)
Theorem 2.4.4 Inverse properties	All the following matrices are square matrices of the same size. 1. I is invertible and $I^{-1} = I$. 2. If A is invertible, so is A^{-1}, and $(A^{-1})^{-1} = A$. 3. If A and B are invertible, so is AB, and $(AB)^{-1} = B^{-1}A^{-1}$. 4. If $A_1, A_2, \ldots, A_k$ are all invertible, so is their product $A_1A_2 \cdots A_k$, and $(A_1 A_2 \ldots A_k)^{-1} = A_k^{-1} \ldots A_2^{-1} A_1^{-1}$ 5. If A is invertible, so is A^k for any $k \geq 1$, and $(A^k)^{-1} = (A^{-1})^k$. 6. If A is invertible and $a \neq 0$ is a number, then aA is invertible and $(aA)^{-1} = \frac{1}{a} A^{-1}$ 7. If A is invertible, so is its transpose A^T, and $(A^T)^{-1} = (A^{-1})^T$.	Note the swapping behaviour in the definition, applying inverses to the matrices inverts their corresponding multiplication order. If multiplying by the inverse returns "1" then it might be similar to multiplying the matrix by 1/A, or dividing A/A? Multiplying by the inverse seems to be the tool for this.
Theorem 2.4.5 Inverse Theorem	The following conditions are equivalent for an n×n matrix A: 1. A is invertible. 2. The homogeneous system $A\mathbf{x} = \mathbf{0}$ has only the trivial solution $\mathbf{x} = 0$. 3. A can be carried to the identity matrix I_n by elementary row operations. 4. The system $A\mathbf{x} = \mathbf{b}$ has at least one solution $\mathbf{x}$ for every choice of column $\mathbf{b}$. 5. There exists an n×n matrix C such that $AC = I_n$.	It needs to be square, no empty vars, and the corresponding matrix that reduces it to I is the inverse. NOTE an identity matrix is essentially in RREF without the constants, it represents 1 for each x in x1, …, xn.

Elementary Matrices

Textbook Ref	Theorem/Definition statement	Notes
Definition 2.12 Elementary Matrices	An n×n matrix E is called an elementary matrix if it can be obtained from the identity matrix I_n by a single elementary row operation (called the operation corresponding to E). We say that E is of type I, II, or III if the operation is of that type: I. Interchanging two rows II. Multiplying one row by a non-zero number III. Adding a multiple of one row to a different row	Elementary row operations in matrix form, convenient application

Theorem 2.5.4 Uniqueness of reduced row-echelon form	If a matrix A is carried to reduced row-echelon matrices R and S by row operations, then R = S.	If you apply row operations to a matrix and get two different reduced forms, those forms must be equal, no matter how you simplify the matrix, it will return the same result.

Linear Transformations

Textbook Ref	Theorem/Definition statement	Notes
Definition 2.13 LU Theorem	A transformation $T: \mathbb{R}^n \to \mathbb{R}^m$ is called a linear transformation if it satisfies the following two conditions for all vectors **x** and **y** in $\mathbb{R}^n$ and all scalars a: T1 $T(\mathbf{x}+\mathbf{y}) = T(\mathbf{x})+T(\mathbf{y})$ T2 $T(a\mathbf{x}) = aT(\mathbf{x})$	Bare in mind the behaviour of linear transformations is essentially matrix multiplication, I'll lean more into this idea of a T() function call passing matrix as a param, this describes how the arithmetic can be reordered.
Theorem 2.6.1 Linearity Theorem	If $T: \mathbb{R}^n \to \mathbb{R}^m$ is a linear transformation, then for each $k = 1, 2, \dots$ $T(a_1\mathbf{x}_1+a_2\mathbf{x}_2+\cdots+a_k\mathbf{x}_k) = a_1T(\mathbf{x}_1)+a_2T(\mathbf{x}_2)+\cdots+a_kT(\mathbf{x}_k)$ for all scalars a_i and all vectors $\mathbf{x}_i$ in $\mathbb{R}^n$.	The effects of multiplying with linearity, scalar multiplication can be applied after a transformation and resolve the same. NOTE continue to actively recognise individual vectors which make up the matrix
Theorem 2.6.4 Rotations	The rotation $R_{-} : \mathbb{R}^2 \to \mathbb{R}^2$ is the linear transformation with matrix $\begin{bmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{bmatrix}$	Think about rotation tools in graphic design software, special matrices like this make vector graphics easily editable
Theorem 2.6.5 Reflections	Let Q_m denote a reflection transformation with respect to the line $y = mx$. Then $Q_m : \mathbb{R}^2 \to \mathbb{R}^2$ is a linear transformation with matrix $\frac{1}{1+m^2}\begin{bmatrix} 1-m^2 & 2m \\ 2m & m^2-1 \end{bmatrix}$	Another graphic design concept, this allows us to easily create mirrored duplicates along an axis/plane
Theorem 2.6.6 Projection	Let P_m be projection on the line $y = mx$. Then $P_m : \mathbb{R}^2 \to \mathbb{R}^2$ is a linear transformation with matrix $\frac{1}{1+m^2}\begin{bmatrix} 1 & m \\ m & m^2 \end{bmatrix}$	This lies at the core of pre-ray-traced graphics and screen-space rendering, determining how 3D objects should appear on a 2D screen from different angles

Summarising your understanding:

- Add notes to the definitions and theorems tables that reflect your understanding of the definition/theorem, how it relates to the learning objectives and things to keep in mind when applying – these notes can be somewhat informal but should be understandable to your OnTrack tutor.

Write a mathematical summary for the module that includes the questions/tasks in the dot points below. Note that the summary should include more than just the questions - we'd recommend having a subheading for each sub-module, and then include the guiding questions where appropriate. You should provide reference to the relevant definitions, theorems and terms in the tables above to help demonstrate your understanding of how the concepts are linked and applied (as you did for Modules 1 and 2).

- Summarise the matrix inversion algorithm in words, referring to an example in your evidence or self-assessment.
- If the rank of an $n \times n$ matrix is <u>less</u> than n, what does this tell us about the inverse matrix? (consider what might happen when applying the inversion algorithm).
- Provide reasoning as to whether or not a non-square matrix could have an inverse.
- What are some strategies for identifying the transformation applied in N3.4.2?

Module 3

Module 3 covers matrix inverses, elementary matrices, and transformations. This module explains how matrices can induce changes on other matrices through multiplication, and how these changes can be reversed through the inverse. Changes induced can range from a single row operation to transformational linear changes upon the entire matrix.

3.1 Matrix Inverses

Essentially the "undoing" of a matrix, inverse matrices $\boldsymbol{A^{-1}}$ return the identity I in multiplication of A * A^{-1}. Not all matrices have inverses. If a change is induced by a matrix through multiplication, the inverse undoes the change induced by the original matrix.

Definitions

Definition 2.11 Inverse Matrix

If multiplying a matrix A by matrix B yields the Identity, A has an inverse (B). Essentially, A^{-1} resets matrix A. ***Since the Identity matrices are square, and $A*A^{-1}$ yields the n*n identity matrix I, the inverse must be square to preserve this structure. Invertible matrices are therefore square.***
This is clarified in **Theorem 2.4.5,** outlining all 5 constraints of n*n matrices according to inverse theorem (see 1: A is invertible).

Uniqueness of inverses
As per **Definition 2.11,** Only a single inverse exists for an invertible matrix. Undoing the effects of a matrix has to be performed in the exact reverse of the original, there's no alternative equal way. **Theorem 2.4.4** provides multiple ways this iterative logic can be reordered, extending to the behaviour of matrix products, where order is key. This goes both ways for inverting and undoing inversion.

Matrix Inversion Algorithm
Prepare an augmented matrix with A on the LHS and its corresponding I on the RHS (A | I), then perform elementary row ops on the whole thing to bring A to I. This will naturally bring I to A^{-1}, but only if an inverse exists for A. ***If any rows in the resulting LHS contain only 0s, the rank of the n*n matrix is less than n, so there's no inverse for A.*** This ties together inverse theorem and inverse properties **(T2.4.5 and T2.4.4)** to achieve sequential, reversible inversion.

Theory

Theorem 2.4.4 Inverse Properties
Multiplying invertible matrices yields an invertible matrix, and Inverting the inverse returns the original matrix. Inverting a matrix product inverts each in reversed order, i.e., $(AB)^{-1} = B^{-1} A^{-1}$.
As per **Definition 2.11**, multiplication must be commutative in regards to I.

Theorem 2.4.5 Inverse Theorem
If the process of Matrix Inversion yields a row of all 0s then A has no inverse. The resulting I must have full rank, otherwise the corresponding A^{-1} isn't possible. Condition 3 largely governs whether the inverse algorithm is successful for a matrix (whether inverse can be found).

3.2 Elementary Matrices
An Elementary Matrix represents a single elementary row operation that can be performed on a corresponding n*n matrix. They're obtained by performing a single row operation on the corresponding identity. This implies that all elementary matrices are square. They express individual row operations in matrix form, and can be applied through multiplication.

Definitions

Definition 2.12
Performing any of the 3 elementary row ops on an I matrix yields an elementary matrix,

which can be applied to another matrix A to perform that same row operation. Elementary matrices are naturally square, and by **Theorem 2.4.5,** invertible.

Theory

Theorem 2.5.4

A sequence of row operations can be represented as an elementary matrix product. Regardless of the specific operations, the resulting RREF will be equal. This directly implies **Definition 2.12** and clarifies that different approaches will yield the same overall transformation when reducing A to RREF.

3.3 Linear Transformations

Transformations are changes induced by another matrix. This allows transitions of one matrix state to another, consistently and predictably, while the original system is preserved. Linear transformations scale the data linearly, preserving these properties and behaviours. These concepts are heavily used in 2D/3D animation/vector graphics.

Definitions

Definition 2.13 LU Theorem

Square matrices can be broken down into simpler sub-parts for lower-left entries and upper-right entries, where lower triangular L and upper triangular U imply that A = LU. This implies transformational decomposition in matrix multiplication, using simpler sequential steps to target specific aspects and effects of a matrix.

Theory

Theorem 2.6.1 Linearity Theorem

For a transformation to be considered linear:

- Vector addition and transformation reordering must yield the same result
- Vector scaling and transformation reordering must also return the same result

As per **Definition 2.13**, matrix composition can express the matrix for any linear transformation.

Theorem 2.6.4 Rotations

Vectors in 2D and 3D space can be rotated by multiplying them by a rotation matrix. Length is preserved while the vector turns around its origin by a specified angle. Rotations don’t distort the data, so the system structure is preserved.

Theorem 2.6.5 Reflections

Reflections flip a vector over a line or plane, resulting in a mirror image. Reflections preserve matrix length and all proportions in the mirrored result, so the structure is always preserved.

Theorem 2.6.6 Projection
Projections impose a vector straight down upon a line or plane, resulting in the closest point on the line/plane to the original vector. This flattens elements/points into simpler forms, like a shadow of a 3D object cast onto a 2D surface.

Determining whether a transformation represents a rotation, or reflection upon a line, can be verified by checking whether the e_{11} and e_{22} values are equal (rotation). If instead, the e_{12} and e_{21} elements are equal, it's a reflection. Both **Theorem 2.6.4** and **2.6.5** define the special matrices responsible for these transformations.

Theorem 2.6.4, 2.6.5, and **2.6.6** reflect constraints outlined in **Theorem 2.4.5**, by which any transformation that reduces the rank of matrix A, has no inverse. **Theorem 2.6.1** expresses linear transformations as a composition of matrix multiplication (matrix A for each transformation T), and building from the compositional approach established in **Definition 2.13**, any linear transformation can be expressed as such.

Module 3 defines all the building blocks for expressing complex transformations as a composition of elementary linear operations. This goes beyond elimination and symbolic problem solving to establish mechanisms for applying, reversing, and decomposing transformations. Where applicable, the inverse can effectively undo any linear transformation induced by the original matrix, which can itself be broken down into individual elementary operations describing the step-by-step linearity of a composition. This scopes row ops within elementary matrices, and their sequence within composition. Multiplying compositions yields more complex transformations, constructed of all the same elementary parts in the same hierarchy.

Reflecting on the content:

- Reflect on what you found interesting/difficult in this module, how it relates to previous content covered in this and other units, and which parts helped you to meet each of the learning objectives. This should be written from your personal perspective and not just read as a general summary of the topic.